

第五章 变换光学和全息照相

§1 衍射系统产生的波前变换

1. 衍射现象

衍射：光的自由传播 $\xrightarrow[\text{限制}]{\text{受到}}$ 振幅 + 相位 $\rightarrow$ 振幅分布改变

无源空间的边值问题

因此：2-D 衍射积分：

$$\tilde{U}(p) = \frac{i}{2\lambda} \oint_{\Sigma} (\cos\theta_0 + \cos\theta) \tilde{U}_0(x) \frac{e^{ikr}}{r} d\Sigma$$

波前函数

Σ ：面上一点

Σ ：衍射屏 / 障碍物

2. 衍射系统屏函数

$\rightarrow$ { 振幅型 各处透射率不同 eg. 光栅
 相位型 各处光程差不同 eg. 透镜

衍射就是波前变换

定义屏函数

$$\tilde{T}(x, y) = \frac{\tilde{U}_2(x, y)}{\tilde{U}_1(x, y)}$$

照明空间 $\rightarrow$ 衍射屏 $\rightarrow$ 衍射空间

$$\tilde{U}_1(x, y) \quad \tilde{U}_2(x, y)$$

$\Sigma(x, y)$ 接收屏

$$\tilde{U}(x, y)$$

光栅： $\tilde{T}(x, y) = \begin{cases} 1 & \text{透光} \\ 0 & \text{不透} \end{cases}$

3. 平面波、球面波的波前函数

1) 平面波

$$\Delta\varphi = \varphi(P) - \varphi(Q) = \vec{k} \cdot \vec{r}$$

$$= k(x \sin\theta_1 + y \sin\theta_2)$$

$$\tilde{U}(x, y) = \tilde{U}(0) \exp[ik(x \sin\theta_1 + y \sin\theta_2)]$$

$$= e^{i[kx \sin\theta_1 + ky \sin\theta_2]}$$

θ_1

$\vec{k} \sin\theta_1$
 $\vec{k} \sin\theta_2$
 $\vec{k}$

$y \sin\theta_2$

整理成：

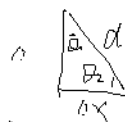
相位差 $(\sin\theta_1, \sin\theta_2)$

空间周期： $d = \frac{\lambda}{\sin^2\theta_1 + \sin^2\theta_2}$

$$a x \sin\theta_1 + a y \sin\theta_2 = \lambda$$

$$\frac{ay}{ax} = \frac{\sin\theta_2}{\sin\theta_1}$$

$$ax (\sin\theta_1 + \frac{\sin^2\theta_2}{\sin\theta_1}) = \lambda$$



$$\frac{ax}{\sin\theta_1} = \frac{\lambda}{\sin^2\theta_1 + \sin^2\theta_2}$$

(2) 球面波

$$\tilde{U}(r) = \left[\frac{A}{r} \right] e^{ikr}, \quad r = \sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2}$$

$$= \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2}} \exp[ik \sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2}] \quad (\text{发散波})$$

$$= \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2}} \exp[-ik \sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2}] \quad (\text{会聚波})$$

4. 傍轴、远场条件

$$\text{当 } \rho_0 = \sqrt{x_0^2 + y_0^2}, \quad \rho = \sqrt{x^2 + y^2} \ll z$$

$$r = \sqrt{(x-x_0)^2 + (y-y_0)^2 + z^2} \approx z \left[1 + \frac{(x-x_0)^2 + (y-y_0)^2}{2z^2} \right]$$

$$= z + \frac{(x-x_0)^2 + (y-y_0)^2}{2z} = z + \frac{\rho_0^2}{2z} + \frac{\rho^2}{2z} - \frac{x x_0 + y y_0}{z}$$

$= r_0$

$$= r_0 + \frac{\rho^2}{2z} - \frac{x x_0 + y y_0}{z}$$

(1) 傍轴条件 (弱)

$$\frac{\rho}{z}, \frac{\rho_0}{z} \ll 1 \Rightarrow \frac{A}{r} \approx \frac{A}{z}$$

$$\tilde{U}(x, y) = \frac{A e^{ikr_0}}{z} \exp[ik \frac{\rho^2}{2z}] \exp[-ik \frac{x x_0 + y y_0}{z}]$$

(2) 远场条件

$$\frac{p^2}{z} \ll \lambda \quad \frac{p^2}{z} \ll \lambda$$

$$r_0 \neq z$$

$$\tilde{u}(x, y) = \frac{A e^{ikz}}{z} \exp\left(ik \frac{p^2}{2z}\right) \exp\left(-ik \frac{xx_0 + yy_0}{z}\right)$$

3. 透镜的相位变换函数

$$\begin{cases} \tilde{U}_\lambda(x, y) = A_\lambda(x, y) \exp[i\varphi_\lambda(x, y)] \\ \tilde{U}_\lambda(x, y) = A_\lambda(x, y) \exp[i\varphi_\lambda(x, y)] \end{cases}$$

$$T(x, y) = K \exp[i\varphi_L(x, y)]$$

玻璃: $k \approx 1$

$$\varphi_L(x, y) = k \cdot [\Delta_1(x, y) + \Delta_2(x, y) + n d(x, y)]$$

$$[d_0 = \Delta_1 + \Delta_2 + d]$$

$$= k [n d_0 - (n-1)(\Delta_1 + \Delta_2)] = C - k(n-1)(\Delta_1 + \Delta_2)$$

r_1, r_2 分别为前后透镜曲率半径 $* r_2 < 0$

$$\Delta_1 = r_1 - \sqrt{r_1^2 - (x^2 + y^2)} \approx \frac{x^2 + y^2}{2r_1}$$

$$\Delta_2 \approx - \frac{x^2 + y^2}{2r_2}$$

$$\varphi_L(x, y) = -k \frac{n-1}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) (x^2 + y^2) = -k \frac{x^2 + y^2}{2f}$$

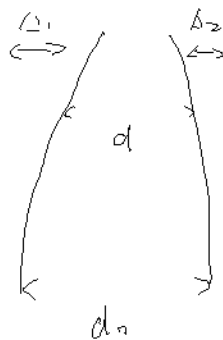
其中 $f = \frac{1}{(n-1)\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$

$$\tilde{T}_L(x, y) = \exp\left(-ik \frac{x^2 + y^2}{2f}\right)$$

取入射波为平面波 $\tilde{U}_\lambda(x, y) = C \cdot \exp(i\varphi_\lambda)$

$$\tilde{U}_\lambda(x, y) = \tilde{U}_\lambda \tilde{T}_L = C \exp\left(-ik \frac{x^2 + y^2}{2f}\right)$$

为会聚中心在 F 处的球面波



取入射波为

$$\vec{U}_i(x, y) = \frac{Ae^{ikz}}{z} \exp\left(ik \frac{x^2+y^2}{2s}\right)$$

$$\vec{U}_a(x, y) = \vec{U}_i \hat{T}_L = \frac{Ae^{ikz}}{z} \exp\left(-ik \frac{x^2+y^2}{2s'}\right)$$

$$\text{其中 } \frac{1}{s'} = \frac{1}{f} - \frac{1}{s}$$

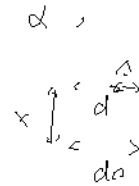
6. 棱镜的相位变换函数

$$\varphi_p(x, y) = k[\Delta(x, y) + nd(x, y)] \quad d + \Delta = d_0$$

$$= k[nd_0 - (n-1)\Delta] = C - k(n-1)\Delta$$

$$\Rightarrow = -k(n-1)\Delta x$$

$$\vec{T}_p = \exp[-ik(n-1)\Delta x]$$



任意取向:

$$\vec{T}_p = \exp[-ik(n-1)(d_1x + d_2y)]$$

$$\text{棱镜法向 } \vec{n} = (\cos\theta_1, \cos\theta_2, \cos\theta_3)$$

$$d_1 = \sin\theta_1, \quad d_2 = \sin\theta_2$$

取入射波为

$$\vec{U}_i(x, y) = \frac{Ae^{ikz}}{z} \exp\left(ik \frac{x^2+y^2}{2s}\right)$$

$$\vec{U}_a(x, y) = \frac{Ae^{ikz}}{z} \exp\left(ik \frac{x^2+y^2}{2s}\right) \exp[-ik(n-1)(d_1x + d_2y)]$$

$$\text{发散中心: } \begin{cases} x_0 = (n-1)d_1s \\ y_0 = (n-1)d_2s \end{cases}$$